

Spiral Model

in

SDLC • Software Development Life Cycle

Presented by:- Kuldeep Sharma
Ayush Kumar Singh
Naveen Pandey
Aman Singh

A Risk-Driven, Iterative Approach to Software Development



Spiral Model

in

SDLC • Software Development Life Cycle

Presented by:- Kuldeep Sharma
Ayush Kumar Singh
Naveen Pandey
Aman Singh

A Risk-Driven, Iterative Approach to Software Development



Agenda

- Introduction
- Key Features
- Spiral Modal Phases
- Why do we use Spiral Modal
- Advantages
- Disadvantages
- Applications
- Conclusion

Introduction

Understanding the Spiral Model

Definition

The Spiral Model is an evolutionary software development process that combines the iterative nature of prototyping with the controlled, systematic aspects of the traditional Waterfall Model. Proposed by Barry Boehm in 1986, it repeats a set of phases in cycles (“spirals”), with each loop refining the product and reducing risk.

Why It Is Used

- Ideal for large, complex, and high-risk projects
- Allows requirements to evolve across iterations
- Combines design + prototyping to manage risk early

Key Features



Risk-Driven

Risk analysis performed in every cycle before proceeding



Iterative

Repeats phases in loops, refining the product each time



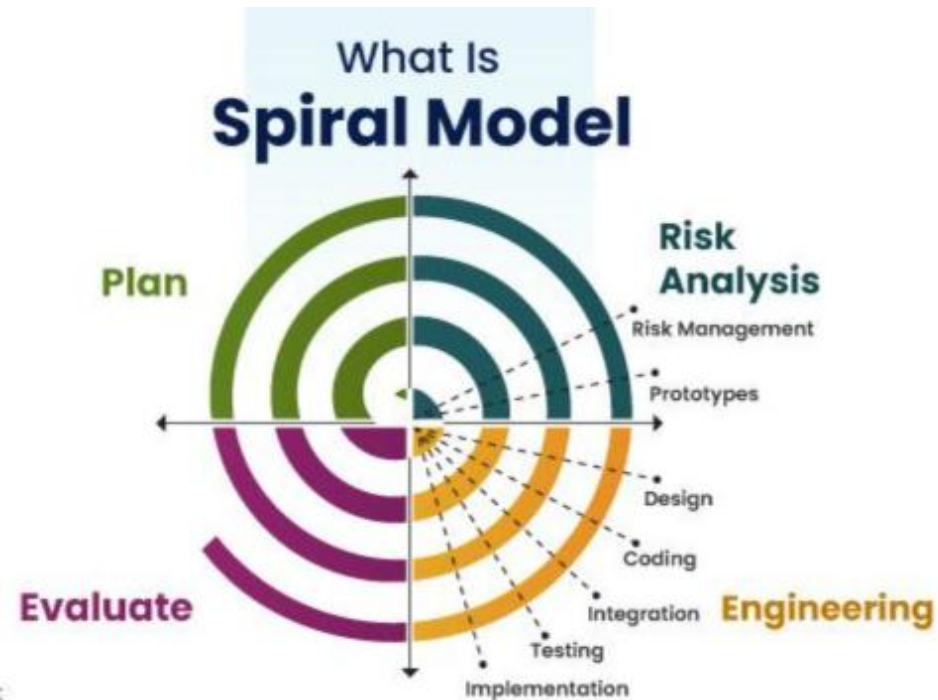
Incremental

Delivers a working prototype at the end of each spiral

Risk-Driven • Iterative SDLC Model

Spiral Model Phases

Each loop of the spiral repeats four key phases



Radius = Cumulative Cost | Angle = Progress



1. Planning

Determine objectives, gather requirements, and identify constraints for the cycle.



2. Risk Analysis

Identify, analyze, and resolve potential risks; build prototypes where needed.



3. Engineering

Develop, code, and test the product increment for the current spiral.



4. Customer Evaluation

Review the output with the customer and plan the next iteration.

1. Planning Phase

Explanation

The Planning phase is the **first stage** of the Spiral Model. In this phase, the project team defines the project goals, gathers requirements, and creates a plan for the current iteration.

Key Points

1. Define the project objectives and goals.
2. Gather and analyze customer requirements.
3. Estimate the project budget, cost, and schedule.
4. Allocate resources such as developers, testers, and tools.
5. Identify project constraints and assumptions.
6. Prepare a detailed plan for the current development cycle.

2. Risk Analysis Phase

Explanation

Risk Analysis is the **most important phase** of the Spiral Model. It focuses on identifying, analyzing, and reducing risks before development begins.

Key Points

1. Identify all possible project risks.
2. Analyze the impact and probability of each risk.
3. Develop strategies to reduce or eliminate risks.
4. Build a prototype if the requirements are unclear.
5. Evaluate technical, financial, and security risks.
6. Resolve major risks before moving to development.

3. Engineering Phase

Explanation

The Engineering phase is where the actual software is designed, developed, and tested based on the planning and risk analysis.

Key Points

1. Design the software architecture and user interface.
2. Develop the software through coding.
3. Implement the planned features of the current iteration.
4. Perform unit testing and integration testing.
5. Fix bugs and improve the software quality.
6. Deliver a working version of the product for evaluation.

4. Customer Evaluation Phase

Explanation

In the Customer Evaluation phase, the developed product is reviewed by the customer to collect feedback and decide improvements for the next iteration.

Key Points

1. Demonstrate the developed software to the customer.
2. Collect customer feedback and suggestions.
3. Verify whether the product meets customer requirements.
4. Identify missing features or required improvements.
5. Plan changes for the next iteration.
6. Obtain customer approval before starting the next cycle.

When to Use the Spiral Model

Choose the spiral approach when a project shows these signs



Requirements Are Unclear

Needs are not well defined and are likely to evolve over time.



High Risk Involved

The project carries significant technical or business risk.



Large, Costly Projects

The system is large in scale with a sizeable budget and timeline.



New / Unproven Technology

A new product line or technology needs careful, staged validation.



Frequent User Feedback Needed

Regular prototypes and customer review are required to steer development.



Long-Term, Evolving Projects

Development spans a long duration where needs may change midway.

Advantages & Disadvantages

Advantages

- ✓ Better risk management through early identification
- ✓ Flexible to requirement changes across iterations
- ✓ Continuous customer feedback improves the final product
- ✓ Well suited for large and complex projects

Disadvantages

- ✗ High cost compared to simpler process models
- ✗ Complex to manage due to many iterative cycles
- ✗ Requires strong risk-analysis expertise to be effective
- ✗ Not suitable for small or low-risk projects

Applications

Where the Spiral Model is commonly adopted



Banking Systems



**E-Commerce
(Amazon, Flipkart)**



Healthcare Systems



Defense & Aerospace



Enterprise Software

The Spiral Model's risk-driven cycles make it well suited to domains where requirements evolve and failure is costly.



Conclusion

- ✓ A risk-driven, iterative model combining prototyping with systematic control
- ✓ Repeats Planning, Risk Analysis, Engineering, and Evaluation in every cycle
- ✓ Best suited for large, complex, and high-risk software projects

Ideal for large, complex & high-risk software projects

Thank You